Savitribai Phule Pune University

TYBCA

Semester – V

(2024 Pattern, with effect from A. Y. 2026-27)

CA- 321 VSEC

Core JAVA Programming

Savitribai Phule Pune University

TYBCA Practical Examination

CA- 321 VSEC: Core JAVA Programming

Duration: 3Hrs. **Max Marks: 35+15=50**

(Q.1) Write a Java program to find addition of Two Numbers using Command Line Arguments. **(10)**

(Q.2) Write a Java program to define a class Book with data members bookId, bookName, and price. Store details of 5 books using an array of objects and display it. **(20)**

OR

(Q.2) Write a java program to define a class person(pid, pname, age, gender). Define default and parameterized constructor. Overload the constructor. Accept the 5 person details and display it. (use this keyword). **(20)**

(Q.3) External Viva **(5)**

(Q.4) Internal Evaluation **(15)**

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TYBCA Practical Examination

CA- 321 VSEC: Core JAVA Programming

Duration: 3Hrs. **Max Marks: 35+15=50**

(Q.1) Write a java program to accept n names of cites from user and display them in descending order. **(10)**

(Q.2) Write a Java program to derive a class Square from class Rectangle. create one more class Circle. Create an interface with only one method called area(). Implement this interface in all classes. Include appropriate data members and constructors in all classes. Write a java program to accept details of Square, Circle & Rectangle and display the area. **(20)**

OR

(Q.2) Write a java program to create a Hash table containing Employee name and Salary. Display the details of the hash table. Also search for a specific Employee and display Salary of that Employee. **(20)**

(Q.3) External Viva **(5)**

(Q.4) Internal Evaluation **(15)**

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TYBCA Practical Examination

CA- 321 VSEC: Core JAVA Programming

Duration: 3Hrs. **Max Marks: 35+15=50**

(Q.1) Write a java program to create a simple applet that displays a message. **(10)**

(Q.2) Write a Java Program which define class Employee with data member as id, name and salary Store the information of n employees and display the name of employee having maximum salary (Use array of object). **(20)**

OR

(Q.2) Write a java program to define an abstract class Staff with members name and address. Define two sub-classes of this class – FullTimeStaff (members - department, salary, hra - 8% of salary, da – 5% of salary) and PartTimeStaff (members - number-of-hours, rate-per-hour). Define appropriate constructors. Write abstract method as calculateSalary() in Staff class. Implement this method in subclasses. Create n objects which could be of either FullTimeStaff or PartTimeStaff class by asking the user‘s choice. Display details of all FullTimeStaff objects and all PartTimeStaff objects along with their salary. **(20)**

(Q.3) External Viva **(5)**

(Q.4) Internal Evaluation **(15)**

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CA- 321 VSEC: Core JAVA Programming

Duration: 3Hrs. **Max Marks: 35+15=50**

(Q.1) Write a java program to check whether a given number is Positive or negative. **(10)**

(Q.2) Write a java program to define a class CricketPlayer(name, no_of_innings, no_of_times_notout, total_runs, bat_avg). Create an array of "n" player objects. Calculate the batting average for each player using a static method avg (). Handle appropriate exception while calculating average. Define static method "sortPlayer" which sort the array on the basis of average. Display the player details in sorted order. **(20)**

OR

(Q.2) Write a package for String operation which has two classes Con and Comp. Con class concatenates two strings, and comp class compares two strings. Also display proper message on execution. **(20)**

(Q.3) External Viva **(5)**

(Q.4) Internal Evaluation **(15)**

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Duration: 3Hrs. **Max Marks: 35+15=50**

(Q.1) Write a java program to display the contents of a file in reverse order. **(10)**

(Q.2) Write a java program to create base class Employee (empcode,empname). Derive the classes anager(designation,club_dues), Scientist (deptname,publications) from Employee class accept the details of n employees and to display the information. **(20)**

OR

(Q.2) Write a java program to create an abstract class Shape with methods area & volume. Derive two classes Sphere (radius), Cylinder (radius, height) from it. Calculate area and volume of both. (Use Method Overriding) **(20)**

(Q.3) External Viva **(5)**

(Q.4) Internal Evaluation **(15)**

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TYBCA Practical Examination

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Duration: 3Hrs. **Max Marks: 35+15=50**

(Q.1) Write a java program to accept username and password if the user name and password are not same, then generate user defined exception "Incorrect Password Exception." **(10)**

(Q.2) Create a package named Series having three different classes to print series: a. Fibonacci series b. Cube of numbers c. Square of numbers Write a java program to generate “n“ terms of the above series. **(20)**

OR

(Q.2) Write a java program that displays the number of characters, lines and words of a file. **(20)**

(Q.3) External Viva **(5)**

(Q.4) Internal Evaluation **(15)**

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TYBCA Practical Examination

CA- 321 VSEC: Core JAVA Programming

Duration: 3Hrs. **Max Marks: 35+15=50**

(Q.1) Write a package game which will have 2 classes Indoor & Outdoor. Use a function display () to generate the list of players for the specific game. Use default & parameterized constructor. **(10)**

(Q.2) Write a java program to define a class MyDate (day, month, year) with methods to accept and display MyDate object. Accept date as dd,mm,yyyy. Throw user defined exception "InvalidDateException" if the date is invalid.
Examples of invalid dates: 12 15 2015, 31 6 1990, 29 2 2001 **(20)**

OR

(Q.2) Write a java program to define a class student having rollno, name and percentage. Define Default and parameterized constructor. Overload the constructor. Accept the 5 student details and display it. (use this keyword). **(20)**

(Q.3) External Viva **(5)**

(Q.4) Internal Evaluation **(15)**

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TYBCA Practical Examination

CA- 321 VSEC: Core JAVA Programming

Duration: 3Hrs. **Max Marks: 35+15=50**

(Q.1) Write a java program to accept names of 'n' cities, insert same into array list collection and display the contents of same array list, also remove all these elements. **(10)**

(Q.2) Write a Java program to create 2 files F1 and F2, Copy the contents of file F1 by changing the case into file F2.F2 = F1 Also copy the contents of F1 and F2 in F3.F3 = F1+F2 Display the contents of F3. **(20)**

OR

(Q.2) Write a Java program to create a class Customer with data members cid, cname, and billAmount. Create an array of 'n' customers and sort them based on bill amount in ascending order. **(20)**

(Q.3) External Viva **(5)**

(Q.4) Internal Evaluation **(15)**

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TYBCA Practical Examination

CA- 321 VSEC: Core JAVA Programming

Duration: 3Hrs. **Max Marks: 35+15=50**

(Q.1) Write a java program to calculate area of Cylinder and Circle. (use super keyword) **(10)**

(Q.2) Write a java program to accept 'n' integers from the user and store them in a collection. Display them in the sorted order. The collection should not accept duplicate elements. (Use a suitable collection). Search for a element using predefined search method in the Collection framework **(20)**

OR

(Q.2) Write a Java program to create a class MyDate with data members dd, mm, and yy. Implement default and parameterized constructors. Display the date in dd-mm-yy format using this keyword. **(20)**

(Q.3) External Viva **(5)**

(Q.4) Internal Evaluation **(15)**

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CA- 321 VSEC: Core JAVA Programming

Duration: 3Hrs. **Max Marks: 35+15=50**

(Q.1) Write a Java program to create class Rectangle having length and breadth. Calculate and display the Area and Perimeter. **(10)**

(Q.2) Write a Java Program to create the hash table that will maintain the mobile number and student name. Display the details of students using enumeration interface. **(20)**

OR

(Q.2) Write a Java program to create a super class Vehicle having members Company and Price. Derive two different classes LightMotorVehicle (mileage) and HeavyMotorVehicle (capacity_in_tons). Accept the information for "n" vehicles and display the information in appropriate form. While taking data, ask user about the type of vehicle first. **(20)**

(Q.3) External Viva **(5)**

(Q.4) Internal Evaluation **(15)**

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Duration: 3Hrs. **Max Marks: 35+15=50**

(Q.1) Write a Java program to accept a number from user and print all prime numbers upto that number (Use Buffered Reader class) **(10)**

(Q.2) Write a program to copy the contents from one file into another file in upper case. **(20)**

OR

(Q.2) Write a Java program to define a class Employee having members – id, name, department, salary. Define default and parameterized constructors. Create a subclass called Worker with private member bonus. Define methods accept and display in both the classes. Create "n" objects of the Worker class and display the details of the worker having the maximum total salary (salary + bonus). **(20)**

(Q.3) External Viva **(5)**

(Q.4) Internal Evaluation **(15)**

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CA- 321 VSEC: Core JAVA Programming

Duration: 3Hrs. **Max Marks: 35+15=50**

(Q.1) Define an Interface Shape with abstract method area(). Write a java program to calculate an area of Circle and Sphere.
(use final keyword) **(10)**

(Q.2) Write an applet program to draw rectangle and oval. **(20)**

OR

(Q.2) Write a Java program to define a class MyNumber having one private integer data member. Write a default constructor initialize it to 0 and another constructor to initialize it to a value. Write methods isNegative, isPositive, isOdd, iseven. Use command line argument to pass a value to the object and perform the above tests. **(20)**

(Q.3) External Viva **(5)**

(Q.4) Internal Evaluation **(15)**

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CA- 321 VSEC: Core JAVA Programming

Duration: 3Hrs. **Max Marks: 35+15=50**

(Q.1) Write a java program to accept a number from the user, if number is Zero then throw user defined exception —Number is 0, otherwise check whether no is prime or not. **(10)**

(Q.2) Write a java program to accept Doctor Name from the user and check whether it is valid or not. (It should not contain digits and special symbol) If it is not valid then throw user defined Exception - Name is Invalid -- otherwise display it. **(20)**

OR

(Q.2) Write a program to accept details of 'n' customers (c_id, cname, address, mobile_no) from user and store it in a file using DataOutputStream class. **(20)**

(Q.3) External Viva **(5)**

(Q.4) Internal Evaluation **(15)**

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(Q.1) Write a java program to accept names of 'n' cities, insert same into array list collection and display the contents of same array list, also remove all these elements. **(10)**

(Q.2) Construct a linked List containing names of colours: red, blue, yellow and orange. Then extend your program to do the following: **(20)**

i. Display the contents of the List using an Iterator
ii. Display the contents of the List in reverse order using a ListIterator
iii. Create another list containing pink and green. Insert the elements of this list between blue and yellow.

OR

(Q.2) Write a program to read the contents of "abc.txt" file, Display the contents of file in uppercase as output. **(20)**

(Q.3) External Viva **(5)**

(Q.4) Internal Evaluation **(15)**

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Duration: 3Hrs. **Max Marks: 35+15=50**

(Q.1) Write a Java program to reverse a number. Accept the number using command line arguments. **(10)**

(Q.2) Create a class circle (member – radius), cylinder (members – radius, height) which implements this interface. Calculate and display the area and volume. **(20)**

OR

(Q.2) Write a java program to create using Runnable interface to display traffic signal. **(20)**

(Q.3) External Viva **(5)**

(Q.4) Internal Evaluation **(15)**

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Duration: 3Hrs. **Max Marks: 35+15=50**

(Q.1) Write a Java program to display Fibonacci series. **(10)**

(Q.2) Write a program to sort HashMap by keys and display the details before sorting and after sorting**.** **(20)**

OR

(Q.2) Write a Java program to define an interface "Operation" which has Methods, area(), volume(). Define a constant PI having a value of 3.142. **(20)**

(Q.3) External Viva **(5)**

(Q.4) Internal Evaluation **(15)**

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Duration: 3Hrs. **Max Marks: 35+15=50**

(Q.1) Define an Interface "Integer" with a abstract method check().Write a Java program to check whether a given number is Positive or Negative. **(10)**

(Q.2) Write a java program to demonstrate Synchronization Example. **(20)**

OR

(Q.2) Write a program to create link list of integer objects. Do the following: **(20)**

i. Add element at first position
ii. Delete last element
iii. Display the size of link list

(Q.3) External Viva **(5)**

(Q.4) Internal Evaluation **(15)**

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(Q.1) Write a Java program to find the sum of elements of an array. Also display the array elements in ascending order. **(10)**

(Q.2) Write a Java Program which defines class Product with data member as id, name and price Store the information of 5products and Display the name of product having minimum price(Use array of object). **(20)**

OR

(Q.2) Write a java program to accept 'N' student names through command line, store them into the appropriate Collection and display them by using Iterator and ListIterator interface. **(20)**

(Q.3) External Viva **(5)**

(Q.4) Internal Evaluation **(15)**

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(Q.1) Define an Interface Shape with abstract method area(). Write a java program to calculate an area of Circle and Sphere. (use final keyword) **(10)**

(Q.2) Write a Java program to store student names and their Roll numbers using an appropriate collection and perform following operations: **(20)**

i. Add a new student and its Roll number (No duplicates)
ii. Remove a student from the collection
iii. Search for a student name and display the roll number

OR

(Q.2) Write a java program to demonstrate Synchronization Example **(20)**

(Q.3) External Viva **(5)**

(Q.4) Internal Evaluation **(15)**

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(Q.1) Write a Java program to accept a number using the BufferedReader class and generate the multiplication table of the given number. **(10)**

(Q.2) Write a java program to show Multiple Threads Execution. **(20)**

OR

(Q.2) Write a Java program to store city names and their STD codes using an appropriate collection and perform following operations: **(20)**

i. Add a new city and its code (No duplicates)
ii. Remove a city from the collection
iii. Search for a city name and display the code

(Q.3) External Viva **(5)**

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OR

(Q.2) Define an interface "Operation" which has methods area(),volume(). Define a constant PI having a value 3.142. Create a class circle (member – radius), cylinder (members – radius, height) which implements this interface. Calculate and display the area and volume. **(20)**

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(Q.2) Write a java program to accept 'N' student names through command line, store them into the appropriate Collection and display them by using Iterator and ListIterator interface. **(20)**

OR

(Q.2) Write a Java program to create LinkedList of String objects and perform the following: **(20)**

i. Add element at the end of the list
ii. Delete first element of the list
iii. Display the contents of list in reverse order

(Q.3) External Viva **(5)**

(Q.4) Internal Evaluation **(15)**

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